

Task-Specific Boundary–Semantic Decoding with Guided Interaction for Remote Sensing Parcel Segmentation

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Abstract

Remote sensing parcel segmentation converts imagery into structured spatial units for land-resource allocation, yield estimation, and spatial decision making. Merged or fragmented regions and inaccurate boundaries distort parcel extent and topology, propagating errors to downstream analyses. Reliable segmentation remains difficult because parcel distributions, scales, geometries, textures, and boundary conditions vary across geographical environments. Meanwhile, predictions must preserve both semantic coherence within parcels and structural separation between parcels. Single-stream decoders provide limited explicit boundary modeling, whereas highly shared multi-task decoders may entangle region prediction and boundary discrimination. Completely isolating the two objectives, however, can weaken their complementary interaction. To coordinate these objectives without forcing them into one decoding representation, we design a boundary–semantic decoding network with guided interaction (BDGI-Net). Task-specific semantic and boundary streams first reconstruct region and structural features separately; a guided mask refinement module (GMRM) then transfers boundary evidence directionally to semantic refinement. An enhanced triplet attention (ETA) mechanism reinforces cross-dimensional responses, while a token aggregation block (TAB) couples local structural details with global context to accommodate heterogeneous textures, scales, and geometries. Experiments on GF-2 and Sentinel-2 datasets show that, relative to the strongest competing result for each metric in each scene, BDGI-Net improves Precision by 0.42–1.44 percentage points and reduces global total classification error (GTC) by 1.15–2.37 percentage points. It attains the best IoU in two scenes and remains within 0.16 percentage points of the best result in the Netherlands. Ablation and synthetic-degradation tests support the component contributions and stability under the specified corruptions. A joint IoU–GTC analysis places BDGI-Net in the Pareto-nondominated set in five of the six original or degraded conditions. These results show that task-specific decoding followed by directional boundary guidance can balance regional accuracy and parcel-level geometric integrity across the evaluated scenes.

Keywords: Remote sensing parcel segmentation, Task-specific boundary–semantic decoding, Guided interaction, Multi-task learning, Semantic segmentation

1. Introduction

Remote sensing parcel segmentation delineates land parcels as structured spatial units for agricultural resource allocation, yield estimation, and spatial decision making [4–6]. If neighboring parcels are erroneously merged, individual parcels are fragmented, or boundaries are omitted, the resulting parcel counts, extents, and topological relations become unreliable. These errors can propagate into downstream analyses that depend on parcel-level geometry, so improving pixel accuracy alone is insufficient when structural integrity is not preserved. Automated delineation is therefore required not only to replace labor-intensive manual mapping at scale, but also to prevent

segmentation errors from undermining the spatial information drawn from growing Earth observation archives [1–3].

Geographical heterogeneity amplifies this difficulty. Across plains, polders, and hilly regions, parcels differ substantially in spatial distribution, scale, geometry, texture, and boundary condition [1, 2, 4]. Region prediction must maintain semantic coherence across these variations, whereas boundary prediction must preserve fine structural transitions that distinguish individual parcels [7, 8]. A representation that favors broad intra-parcel consistency can suppress local boundary responses, while one that overemphasizes fine structure can fragment otherwise coherent regions. Thus, heterogeneous scenes require coordination of two related but non-identical demands: regional completeness and boundary separation.

Existing parcel segmentation architectures approach this coordination problem through two main routes, as illustrated in Fig. 1. Single-stream networks (Fig. 1(a)) optimize region prediction within a unified decoder; because parcel boundaries

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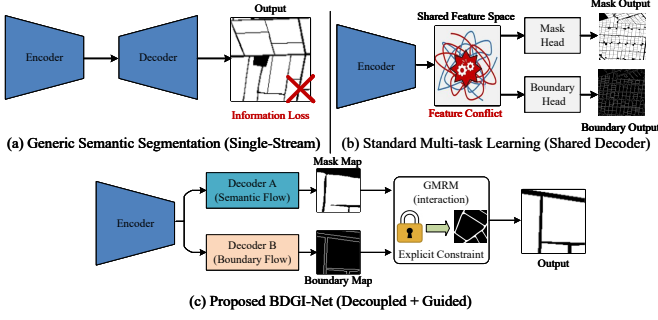


Figure 1: Comparison of three parcel segmentation paradigms according to the decoding strategy: (a) a generic single-stream structure with implicit boundary learning, (b) a standard multi-task architecture with coupled optimization through a shared decoder, and (c) the proposed BDGI-Net with task-specific boundary-semantic decoding and directional boundary guidance.

are represented only implicitly, their responses can be weakened during progressive semantic abstraction. Multi-task networks (Fig. 1(b)) add boundary supervision, but highly shared or coupled decoding representations still force region consistency and boundary discrimination to compete within the same feature space. Excessive sharing can therefore introduce cross-task interference. The opposite strategy of fully isolating the two objectives is also incomplete, because semantic regions and their enclosing boundaries contain complementary information that must be exchanged to correct ambiguous predictions. How to combine task-specific representation learning with controlled complementary interaction consequently remains an open issue in heterogeneous remote sensing parcel segmentation [10, 13, 14].

To address this open issue, we design BDGI-Net to reconstruct task-specific features before transferring boundary guidance directionally, as shown in Fig. 1(c). Because semantic regions and parcel boundaries require different representations, BDGI-Net first assigns them to separate decoding streams, reducing direct competition during feature reconstruction. Because strict separation would suppress useful cross-task cues, GMRM then converts boundary features into spatial guidance for semantic refinement without restoring unrestricted feature sharing. Finally, because geographical heterogeneity produces large variations in texture, scale, and geometry, ETA strengthens cross-dimensional feature interaction and TAB combines local structural detail with global context. This ordered decode-then-guide pipeline links each design choice to a specific failure mode instead of treating boundary learning as an auxiliary output added to a conventional decoder.

The main contributions are summarized as follows:

- *Task-specific boundary-semantic decoding.*

Shared decoding can force the different reconstruction requirements of region prediction and boundary discrimination into one feature space. We therefore develop parallel semantic and boundary decoding streams with a shared encoder, allowing each task to recover the features it requires while reducing direct competition between regional coherence and fine structural delineation.

- *Controlled boundary-guided interaction.*

Although task-specific decoding reduces interference, complete separation weakens the complementary relation between a parcel and its boundary. We therefore develop GMRM to transform boundary features into spatial guidance for semantic refinement, helping suppress erroneous connections and preserve parcel-level structural consistency without reverting to unrestricted feature fusion.

- *Hierarchical reinforcement for geographical heterogeneity.*

Variations in parcel texture, scale, and geometry require both stable cross-dimensional responses and context across spatial ranges. We therefore combine ETA with TAB: ETA augments cross-dimensional attention with global channel context, whereas TAB links windowed detail and global context to preserve complete regions and boundaries.

The remainder of this paper is organized as follows. Section 2 reviews related work. Section 3 presents the proposed method. Section 4 describes the experimental settings. Section 5 reports and analyzes the experimental results. Section 6 discusses the implications and next-stage development of the proposed design. Finally, Section 7 concludes this study.

2. Related Work

Existing deep learning approaches to remote sensing parcel segmentation can be distinguished by where and how they represent structural information: edge detection methods predict boundaries directly, single-task networks optimize parcel regions, and multi-task frameworks couple region and boundary objectives. This organization exposes a progressive bottleneck. Explicit edges provide structure without sufficient region semantics, region-only prediction provides coherence without explicit separation, and coupled multi-task learning can still entangle the two information demands.

2.1. Edge Detection Methods

Edge detection methods treat parcel delineation as a local structural prediction problem. RCF [23] integrates multi-level convolutional features to strengthen boundary responses beyond traditional operators such as Canny [24], while DexiNed [22] uses dense connections and hierarchical multi-scale learning to improve boundary continuity [21, 22]. These designs are effective when parcel transitions produce observable local gradients.

Their limitation follows from the same local emphasis. Because edge responses contain limited high-level region semantics, internal texture changes or gradual land-cover transitions can be mistaken for parcel borders, whereas weak or discontinuous structural cues can be missed [25]. Edge evidence alone therefore cannot determine whether locally detected fragments enclose a complete parcel, restricting its ability to preserve region completeness and continuous separation simultaneously.

2.2. Single-task Learning Networks

Single-task semantic segmentation instead optimizes coherent parcel regions through end-to-end pixel-wise prediction. U-Net [26] combines shallow spatial details with deep semantics through skip connections, and DeepLabV3+ [27] enlarges the receptive field through atrous spatial pyramid pooling. Their unified objectives support region-level consistency and make parcel mask generation more direct.

However, because the optimization target is dominated by regional classification, fine boundary cues are represented only implicitly and may decay during semantic abstraction. This bottleneck can produce boundary discontinuities, geometric distortions, or connections between neighboring parcels even when the predicted region appears locally coherent [28]. Increasing semantic capacity alone does not explicitly resolve this structural ambiguity; a mechanism that distinguishes region and boundary information is required.

2.3. Multi-task Learning Networks

Multi-task learning introduces boundary-related supervision to compensate for the structural weakness of region-only prediction. Psi-Net [16] showed that auxiliary contour-related tasks can support a primary segmentation objective. Building on this principle, BsiNet [15] jointly predicts parcel masks, boundaries, and distance maps; SEANet [17] couples mask prediction with semantic edge learning; and REAUNet [18] uses refined edge-aware convolutions to recover boundary detail. HBGNet [19] further combines semantic extraction, auxiliary boundary mining, and local-global context to represent parcels with varying sizes and shapes.

Boundary supervision addresses the absence of explicit structure, but it does not by itself determine how the tasks should share information. When semantic and boundary predictions depend on a highly shared decoder or direct feature fusion, their different optimization requirements can interfere within the same representation. Conversely, fully independent streams cannot exploit the complementary relation between parcel interiors and their enclosing boundaries. Recent vectorization-oriented methods make this distinction especially relevant: RBP-MTL separates region, boundary, and parcel objectives [10], while CPVF couples boundary and parcel prediction for vector output [20]. These methods target parcel vectorization, whereas the present study focuses on raster segmentation and asks how task-specific mask and boundary features should interact. BDGI-Net therefore separates their decoding paths first and then uses boundary features to guide, rather than indiscriminately mix with, semantic refinement.

3. Methodology

This section presents the overall architecture of BDGI-Net and describes its main components, including hierarchical feature enhancement, task-specific semantic and boundary decoding, guided mask refinement, and the optimization objective.

3.1. Architecture of BDGI-Net

BDGI-Net follows a two-stage rationale: task-specific decoding first reduces representation interference, after which directional boundary guidance restores the complementarity required for semantic refinement. Before decoding, hierarchical features are reinforced across dimensions and spatial ranges to accommodate geographical variations in parcel texture, scale, and geometry.

As shown in Fig. 2, a multi-scale encoder extracts hierarchical features from the input image $\mathbf{I} \in \mathbb{R}^{C \times H \times W}$, where C , H , and W denote the number of channels, height, and width, respectively. We denote these features by $\mathbf{F}_i \in \mathbb{R}^{C_i \times (H/2^{i-1}) \times (W/2^{i-1})}$, where $C_i \in \{32, 64, 128, 256, 512\}$ and $i \in \{1, 2, 3, 4, 5\}$. ETA refines $\mathbf{F}_1$ – $\mathbf{F}_4$ to strengthen cross-dimensional interactions in spatially detailed features. TAB processes $\mathbf{F}_5$ to combine local structure with the global context required to represent complete parcels.

The reinforced hierarchy enters two task-specific streams. The semantic decoder reconstructs parcel masks through multi-scale fusion and encoder skip connections, whereas the boundary decoder recovers fine structural transitions. GMRM subsequently converts the boundary representation into spatial guidance for semantic refinement. The resulting data flow first preserves task-specific decoding paths and then restores their complementary interaction in one direction.

3.2. Hierarchical Feature Reinforcement

Geographical heterogeneity requires the feature hierarchy to preserve fine structures and the wider context of complete regions. We therefore apply ETA to low- and mid-level encoder features, where spatial-channel interactions carry structural detail, and TAB to the highest-level feature, where local-global context supports subsequent reconstruction.

3.2.1. Enhanced Triplet Attention (ETA)

Parcel structures are expressed jointly through spatial layout and channel responses, so modeling either dimension independently can leave their dependencies underused. Standard triplet attention addresses this problem by permuting the feature tensor and learning interactions across the (C, H) , (C, W) , and (H, W) views [30]. However, its three branches do not explicitly encode global context. In large homogeneous or weakly textured parcels, this limitation can make local responses unstable and fragment otherwise coherent regions.

Because cross-dimensional interaction and global consistency are both required, we develop ETA by retaining the three rotation branches and adding a parallel global context branch, as shown in Fig. 3. The rotation branches preserve spatial-channel dependencies, while the additional branch supplies an image-level channel descriptor that stabilizes feature responses within parcels.

Specifically, we define $\mathbf{F}_k$ ($k \in \{1, 2, 3\}$) as the original (C, H, W) arrangement and its permuted forms (H, C, W) and (W, H, C) . These permutations allow the cross-dimensional gating operation to model dependencies in the (H, W) , (C, W) ,

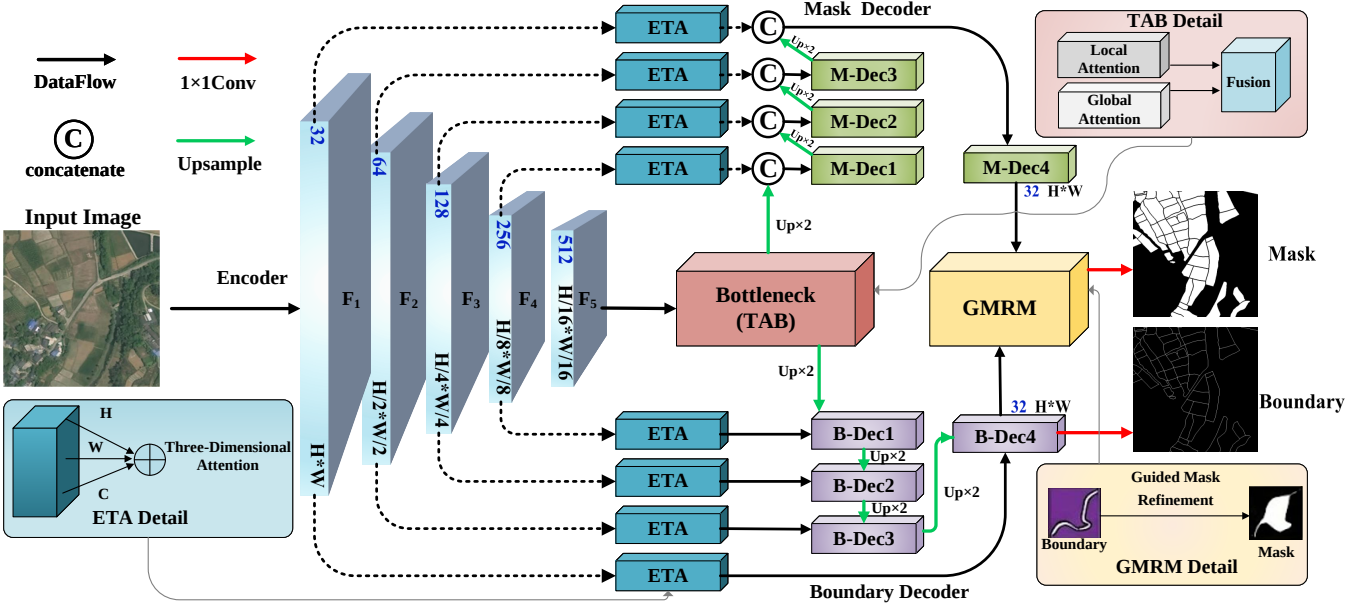


Figure 2: Overall architecture of BDGI-Net. A shared encoder extracts hierarchical features F_1 – F_5 , which are reinforced by ETA and TAB before entering task-specific semantic and boundary decoders. GMRM transfers controlled boundary guidance to semantic refinement. The symbol $\odot$ denotes channel-wise concatenation, and C , H , and W denote channel number, height, and width, respectively.

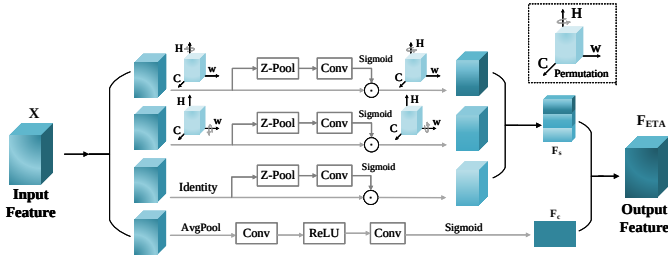


Figure 3: The structure of the enhanced triplet attention (ETA) module. Here, $\odot$ denotes the broadcast Hadamard product used for element-wise multiplication of feature maps.

and (H, C) views, respectively. For each view, we define the gating function as

$$\Psi(\mathbf{X}) = \sigma(\Phi_{7 \times 7}(\text{Z-Pool}(\mathbf{X}))), \quad (1)$$

where Z-Pool concatenates the maximum and average responses along the first feature dimension, $\Phi_{7 \times 7}$ denotes a 7×7 convolution followed by batch normalization, and σ is the sigmoid function. We apply the resulting attention map element-wise to the corresponding feature representation so that each view retains only the responses supported by its cross-dimensional context.

We inverse-permute and average the outputs of the three cross-dimensional branches to obtain

$$\mathbf{F}_s = \frac{1}{3} \sum_{k=1}^3 \text{Permute}_k^{-1}(\Psi(\mathbf{F}_k) \odot \mathbf{F}_k), \quad (2)$$

where $\odot$ denotes the broadcast Hadamard product.

In addition to the three cross-dimensional branches, we design a global context branch for ETA. We compress the input

feature by global average pooling and process it with two 1×1 convolutions and a nonlinear activation to obtain a channel-wise descriptor $\mathbf{F}_c$. We then combine this descriptor with the cross-dimensional feature so that image-level channel evidence can stabilize locally ambiguous responses:

$$\mathbf{F}_{\text{ETA}} = \frac{1}{2}(\mathbf{F}_s + \mathbf{F}_c). \quad (3)$$

Because the global descriptor complements rather than replaces the rotated views, ETA preserves cross-dimensional structural interactions while stabilizing parcel interiors.

3.2.2. Token Aggregation Block (TAB)

Fine parcel boundaries are best represented from local neighborhoods, whereas complete parcel regions require dependencies that may span distant positions. Restricting attention to local windows can therefore miss long-range semantic continuity, while relying only on global attention can dilute fine structural detail. Because neither scale alone satisfies both requirements, we design TAB as a parallel local-global attention block. Its Window MSA branch preserves local geometric patterns, and its Global MSA branch captures long-range contextual dependencies; their aggregation provides the semantic decoder with global support without discarding the structures required by the boundary decoder.

Overall Architecture. As shown in Fig. 4(a), TAB adopts a parallel local-global attention structure. Given the input feature $\mathbf{F}_{\text{in}}$, we first rearrange its dimensions from (B, C, H, W) to (B, H, W, C) and apply Layer Normalization. We then process the normalized feature through two parallel branches: a global multi-head self-attention (Global MSA) branch and a window-based multi-head self-attention (Window MSA) branch.

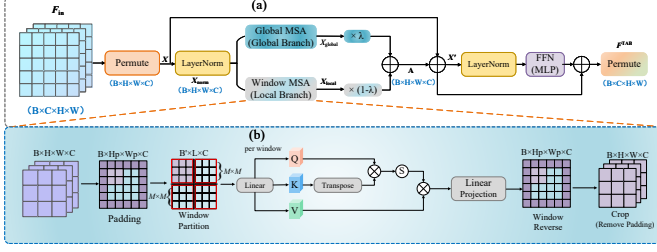


Figure 4: The architecture of the token aggregation block (TAB). Here, $\otimes$ denotes batched matrix multiplication, $\oplus$ denotes element-wise addition, and the circle with S represents the softmax operation applied to the attention scores.

Local-Global Interaction. The Global MSA branch operates over the entire spatial domain to capture long-range dependencies and provide global contextual information. In contrast, the Window MSA branch focuses on local structural details. As shown in Fig. 4(b), we divide the feature map into non-overlapping 7×7 windows, apply padding and cropping when necessary, and compute self-attention independently within each window. This construction limits local-attention cost while keeping the neighborhood scale fixed across scenes.

To prevent either spatial range from becoming the sole reconstruction basis, we combine the two branch outputs using the following weighted aggregation:

$$A = (1 - \lambda)X_{local} + \lambda X_{global}, \quad (4)$$

where λ controls the contribution of global contextual information; we set $\lambda = 0.5$ in our implementation.

We then add the aggregated attention feature to the rearranged input through a residual connection:

$$X' = X + A. \quad (5)$$

Finally, we apply a feed-forward network with Layer Normalization and residual learning to transform the aggregated context while preserving the incoming representation:

$$X_{out} = X' + \text{FFN}(\text{LN}(X')). \quad (6)$$

We subsequently rearrange the output back to (B, C, H, W) to obtain F_{TAB} . By preserving complementary spatial ranges, TAB reinforces long-range region continuity without discarding fine parcel structures.

3.3. Dual-decoder Architecture with Guided Mask Refinement

The design addresses the representation tension in two ordered steps: task-specific decoding reduces direct cross-task competition, and guided mask refinement reintroduces boundary guidance to the semantic stream.

3.3.1. Task-Specific Decoding Streams

Semantic region prediction depends on spatial coherence and parcel extent, whereas boundary delineation depends on localized geometric transitions. When both objectives are reconstructed in the same decoding space, gradients that favor

smooth region interiors can suppress boundary-sensitive responses, while gradients that emphasize thin structures can destabilize region completeness. We therefore use two task-specific decoding streams rather than increasing the capacity of a shared decoder.

The semantic decoder progressively recovers spatial resolution through hierarchical fusion and encoder skip connections, preserving coherent parcel regions. In parallel, the boundary decoder integrates hierarchical features to recover fine structural cues. Both streams retain a shared encoder because they originate from the same image, while separate decoders prevent their reconstruction objectives from competing directly.

3.3.2. Guided Mask Refinement Module (GMRM)

Task-specific decoding reduces direct representation competition but creates a second problem: independently reconstructed semantic features cannot directly use boundary evidence to correct uncertain parcel transitions. This limitation is particularly harmful when local semantic responses connect regions that should remain separate. Because the boundary stream already encodes the structural cues required at these locations, we design GMRM to transfer them directionally into semantic refinement rather than returning to unrestricted bidirectional fusion [29].

As shown in Fig. 5, GMRM follows a guidance-by-constraint principle. It converts the boundary feature into spatial attention for mask modulation, recalibrates the boundary representation, and then performs additive fusion. This ordering constrains semantic correction with boundary evidence while preserving the task-specific roles of the two decoders.

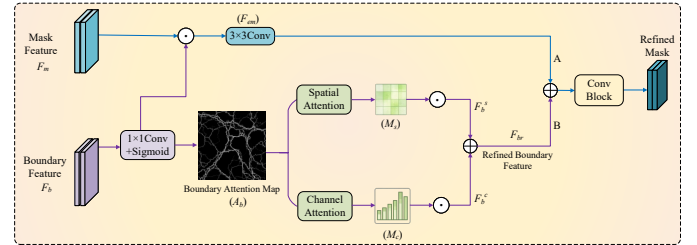


Figure 5: The structure of the guided mask refinement module (GMRM). The module refines semantic mask features by incorporating boundary cues from the boundary decoding stream. Here, $\odot$ denotes element-wise multiplication and $\oplus$ denotes element-wise addition.

Fig. 5 illustrates the interaction between boundary and semantic features in GMRM. Given the boundary feature F_b , we apply a 1×1 convolution followed by a sigmoid activation to generate a boundary attention map A_b . We then use this map to modulate the input mask feature F_m spatially:

$$F_g = F_m \odot A_b, \quad (7)$$

where $\odot$ denotes element-wise multiplication. We subsequently apply a 3×3 convolution to the guided feature to obtain the enhanced mask feature F_{em} .

Meanwhile, we refine the boundary feature F_b through spatial and channel attention because reliable guidance requires both boundary location and channel relevance. The spatial branch

aggregates channel responses and applies a 7×7 convolution followed by sigmoid activation, while the channel branch uses global average pooling and channel-wise recalibration in a manner similar to Squeeze-and-Excitation [32]. We combine the two refined boundary features as

$$\mathbf{F}_{br} = \mathbf{F}_b^s + \mathbf{F}_b^c, \quad (8)$$

where $\mathbf{F}_b^s$ and $\mathbf{F}_b^c$ denote the spatially and channel-wise refined boundary features, respectively.

Finally, we use additive fusion to integrate the enhanced mask feature and the refined boundary feature without introducing a second shared decoder:

$$\hat{\mathbf{F}}_m = \mathcal{R}(\mathbf{F}_{em} \oplus \mathbf{F}_{br}), \quad (9)$$

where $\oplus$ denotes element-wise addition and $\mathcal{R}(\cdot)$ represents the final convolutional refinement block. We design this guided fusion to inject boundary information into the semantic stream and reduce erroneous connections between adjacent parcels.

Because GMRM injects boundary evidence only after task-specific decoding, it restores complementarity without recreating a fully shared representation. The correction targets parcel transitions, where erroneous pixel connections cause under-segmentation and topological distortion.

3.4. Loss Function

The two decoders predict outputs with different spatial characteristics, so they require supervision matched to their roles. The semantic mask covers complete parcel regions and is sensitive to foreground-background imbalance; its objective therefore combines pixel-wise binary cross entropy (BCE) with overlap-oriented Dice loss. The boundary output contains thin structural targets, for which BCE supplies direct pixel-level supervision. We weight the two losses to keep semantics primary and boundary learning complementary.

3.4.1. Mask Loss

We formulate mask prediction as pixel-level classification, but BCE alone can be dominated by the background when foreground parcels occupy a smaller image proportion. Because accurate parcel extraction also depends on the overlap of the predicted and reference regions, Dice loss complements BCE with a region-level objective.

Let $y_i \in \{0, 1\}$ denote the label of the i -th pixel and $\hat{y}_i$ the corresponding prediction. The BCE loss is written as

$$\mathcal{L}_{bce} = -\frac{1}{N} \sum_{i=1}^N \left[y_i \log(\hat{y}_i) + (1 - y_i) \log(1 - \hat{y}_i) \right]. \quad (10)$$

With ϵ as a smoothing term to avoid division by zero (e.g., 1×10^{-7}), we define the Dice loss as

$$\mathcal{L}_{dice} = 1 - \frac{2 \sum_{i=1}^N (\hat{y}_i y_i) + \epsilon}{\sum_{i=1}^N \hat{y}_i + \sum_{i=1}^N y_i + \epsilon}, \quad (11)$$

The resulting mask loss combines the BCE and Dice terms: BCE constrains pixel-wise classification, whereas Dice loss reinforces parcel-region overlap.

3.4.2. Boundary Loss

Boundary prediction targets thin structures for which direct pixel-wise discrimination is required. We therefore train the boundary branch with BCE, which measures the difference between the predicted boundary map and its reference and supplies explicit supervision to the task-specific boundary decoder.

3.4.3. Total Loss

The final loss combines the two tasks:

$$\mathcal{L}_{total} = \alpha \mathcal{L}_{mask} + (1 - \alpha) \mathcal{L}_{boundary}, \quad (12)$$

where α and $1 - \alpha$ balance the two branches. Because mask prediction is the primary task and boundary learning provides complementary structural supervision, this weighting controls their optimization contributions without forcing them into a shared decoding representation.

4. Experimental Setting

In this section, we provide a detailed discussion of the dataset, training settings, and evaluation metrics.

4.1. Datasets

To evaluate whether the same interaction principle remains useful across distinct sensors, resolutions, and parcel morphologies, we construct the training and evaluation corpus from multi-source remote sensing data. It covers high-resolution GaoFen-2 panchromatic-multispectral fusion imagery and medium-resolution Sentinel-2 multispectral imagery. We use GaoFen-2 imagery with a 1-m spatial resolution for Shandong (10 661 $\times$ 8 769 pixels) and Sichuan (144 768 $\times$ 145 883 pixels), and Sentinel-2 Level-2A imagery with a 10-m spatial resolution for the Netherlands (42 845 $\times$ 31 580 pixels). The reference parcels come from manually annotated farmland vectors for the Chinese regions and cadastral vectors from the Netherlands National Geospatial Portal (<https://www.pdok.nl/>).

The three study regions are representative in terms of parcel morphology and scene complexity. Shandong, located in northern China, features regularly shaped parcels that are almost entirely rectangular with clearly defined boundaries. The Netherlands area is primarily composed of regularly arranged farmland with dense distribution, where partially blurred boundaries are common. Sichuan, situated in southwestern China, is dominated by hilly terrain. While farmland in the plains is relatively regular and rectangular, parcels in the hilly regions are narrow and complex in shape. Moreover, the presence of dense vegetation and forests frequently results in adjacent parcel boundaries appearing blurred or fragmented. We cropped the original imagery into 512 $\times$ 512-pixel patches using a sliding window with a stride of 256 pixels and 50% overlap. Rotation and scaling increased the appearance variation of the cropped samples. This process produced 2 080, 4 250, and 3 640 samples for Sichuan, Shandong, and the Netherlands, respectively, as summarized in Table 1.

Table 1: Patch counts before and after geometric expansion. Orig. and Exp. denote the original and expanded patch totals, respectively.

Dataset	Area	Training data (Orig./Exp.)	Test data (Orig./Exp.)
GF-2 dataset	Shandong (China)	850/3,400	212/850
	Sichuan (China)	416/1,664	104/416
Sentinel-2 dataset	Netherlands	728/2,912	182/728

4.2. Training Settings

We set the mask-to-boundary loss-weight ratio to 6:4, as evaluated in Section 5.1. We divided the dataset into training and test subsets at a ratio of 0.8:0.2 and reserved 10% of the training subset for validation. We trained each model for 150 epochs with Adam, an initial learning rate of 10^{-4} , and a batch size of 4. We implemented all experiments in PyTorch and ran them on one NVIDIA GeForce RTX 4080 GPU with 16 GB of memory.

4.3. Evaluation Indicators

We use Precision (P) and Intersection over Union (IoU) as pixel-based evaluation metrics [33]. Precision measures the proportion of correctly predicted positive pixels among all pixels predicted as positive, and we compute it as follows:

$$P = \frac{TP}{TP + FP}, \quad (13)$$

We use IoU to assess pixel-level segmentation accuracy by dividing the overlap between predicted and reference parcels by their union:

$$IoU = \frac{TP}{TP + FP + FN}, \quad (14)$$

where TP, FP, TN, and FN denote the true positive, false positive, true negative, and false negative pixels, respectively.

To further evaluate the geometric accuracy of the extracted parcels, we use object-based classification error metrics [34]. Assume there are m predicted parcels, and we match each predicted parcel M_i to the reference parcel O_i with which it has the largest overlapping area. Based on $S(M_i)$, $S(O_i)$, and $S(M_i \cap O_i)$, we define the errors as follows:

Given the denominator in Eq. 15, over-classification error (OC) is zero when the matched prediction fully covers the reference parcel and increases as reference coverage decreases:

$$OC(M_i) = 1 - \frac{S(M_i \cap O_i)}{S(O_i)}. \quad (15)$$

Given the denominator in Eq. 16, under-classification error (UC) is zero when the matched prediction stays within the reference parcel and increases with unsupported predicted area:

$$UC(M_i) = 1 - \frac{S(M_i \cap O_i)}{S(M_i)}. \quad (16)$$

These interpretations follow directly from the denominators: if $M_i \subset O_i$, then $UC(M_i) = 0$ and $OC(M_i) > 0$; if $O_i \subset M_i$, then $OC(M_i) = 0$ and $UC(M_i) > 0$.

We compute total classification error (TC) to jointly evaluate over-classification and under-classification:

$$TC(M_i) = \sqrt{\frac{OC(M_i)^2 + UC(M_i)^2}{2}}, \quad (17)$$

Based on these per-object errors, we use area-weighted averaging to quantify overall geometric deviation. We define global over-classification error (GOC), global under-classification error (GUC), and global total classification error (GTC), respectively, as follows:

$$GOC = \sum_{i=1}^m \left(OC(M_i) \times \frac{S(M_i)}{\sum_{j=1}^m S(M_j)} \right), \quad (18)$$

$$GUC = \sum_{i=1}^m \left(UC(M_i) \times \frac{S(M_i)}{\sum_{j=1}^m S(M_j)} \right), \quad (19)$$

$$GTC = \sum_{i=1}^m \left(TC(M_i) \times \frac{S(M_i)}{\sum_{j=1}^m S(M_j)} \right), \quad (20)$$

where the weight is the proportion of each parcel's area to the total predicted area, ensuring that larger parcels contribute more significantly to the overall error. The combined use of IoU and the global classification error metrics enables a comprehensive assessment of the model performance from both pixel-level accuracy and object-level geometric integrity perspectives.

5. Experimental Results

5.1. Effect of Task Loss Weights

Table 2: Performance of BDGI-Net under different task loss weights, where α and $1 - \alpha$ denote the weights of the mask and boundary losses, respectively.

α	Precision $\uparrow$	IoU $\uparrow$	GOC $\downarrow$	GUC $\downarrow$	GTC $\downarrow$
0.5	0.8825	0.8213	0.1892	0.3709	0.3452
0.6	0.9049	0.8225	0.2555	0.2859	0.3284
0.7	0.8904	0.8197	0.2199	0.3303	0.3353
0.8	0.8833	0.8219	0.1659	0.3796	0.3367
0.9	0.8741	0.8187	0.1775	0.3871	0.3478

To determine whether the mask and boundary objectives remain complementary rather than one suppressing the other, we evaluate task-weight sensitivity on the Netherlands dataset, whose dense and morphologically heterogeneous parcels challenge both region prediction and boundary delineation. Excessive semantic weighting may weaken structural supervision, whereas excessive boundary weighting may limit region learning. Because mask prediction is the primary task, we vary its loss weight α from 0.5 to 0.9 and assign $1 - \alpha$ to the auxiliary boundary loss. This controlled sweep keeps the architecture and data fixed, so changes in regional and geometric metrics can be attributed to the optimization balance.

As shown in Fig. 6 and Table 2, the balanced setting ($\alpha = 0.5$) does not yield the best overall performance. Increasing the mask

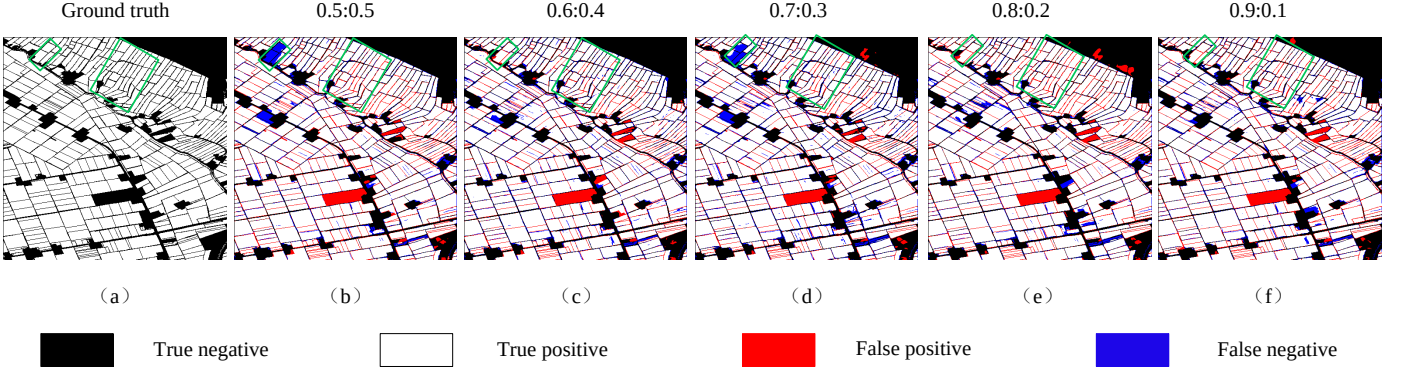


Figure 6: Performance variation of BDGI-Net under different task loss weights. Black, white, red, and blue regions in the error maps denote true negatives, true positives, false positives, and false negatives, respectively.

weight to $\alpha = 0.6$ produces the highest Precision (0.9049) and IoU (0.8225), together with the lowest GUC (0.2859) and GTC (0.3284). Although $\alpha = 0.8$ produces the lowest GOC, its Precision and IoU are lower. These results show that a slightly larger mask weight improves region prediction while retaining sufficient boundary supervision.

Performance generally declines when α exceeds 0.6. The accompanying increase in GUC indicates weaker separation as boundary supervision is reduced. At $\alpha = 0.9$, Precision and IoU decrease to 0.8741 and 0.8187, while GUC and GTC increase to 0.3871 and 0.3478. Thus, prioritizing mask learning too strongly compromises the geometric benefit of the auxiliary task. Among the five supplied point-estimate settings, $\alpha = 0.6$ is the only Pareto-nondominated choice on higher IoU and lower GTC; it weakly improves both objectives relative to every alternative. This descriptive two-objective check avoids selecting the weight through an unreported composite score. We therefore use $\alpha = 0.6$ in all subsequent experiments.

5.2. Comparison with Other Networks

To test whether the ordered decode-then-guide principle remains effective across regular plains, dense polders, and irregular hilly parcels, we compare BDGI-Net on all three scenes against region-only, boundary-aware, and multi-task alternatives. This baseline coverage distinguishes gains from a stronger generic segmenter, explicit boundary supervision, or a different feature-sharing strategy. We evaluate every method with both pixel-level overlap and parcel-level geometric errors because either family alone can conceal merging, fragmentation, or omission.

We compare BDGI-Net with five parcel-oriented networks and one general segmentation baseline. SEANet [17] uses semantic edge-aware multi-task learning, while BsiNet [15] jointly predicts masks, boundaries, and distance maps. REAUNet [18] employs refined edge-aware convolutions, and HBGNet [19] integrates semantic extraction, auxiliary boundary mining, and local-global context. TFNet [35] combines Segment Anything Model (SAM) boundary priors [36] with high-level semantics. DeepLabV3+ [27] provides a widely used semantic segmentation baseline. Together, these methods cover region-only,

Table 3: Performance comparison of BDGI-Net with other networks.

Scene	Network	Precision $\uparrow$	IoU $\uparrow$	GOC $\downarrow$	GUC $\downarrow$	GTC $\downarrow$
Shandong	BDGI-Net	0.9274	0.8306	0.1989	0.1797	0.2445
	HBGNet	0.9220	0.8288	0.1295	0.2692	0.2723
	TFNet	0.9087	0.8264	0.1072	0.3438	0.2941
	REAUNet	0.8957	0.8227	0.1131	0.3755	0.3215
	SEANet	0.9070	0.8204	0.1186	0.2878	0.2615
	BsiNet	0.9043	0.8122	0.1513	0.3060	0.2912
	DeepLabV3+	0.9014	0.8013	0.1073	0.3589	0.3038
Netherlands	BDGI-Net	0.9049	0.8225	0.2555	0.2859	0.3284
	HBGNet	0.9007	0.8241	0.2702	0.2988	0.3544
	TFNet	0.8677	0.7756	0.6267	0.4009	0.5842
	REAUNet	0.8785	0.8013	0.1276	0.4337	0.3533
	SEANet	0.8761	0.8129	0.1472	0.4200	0.3521
	BsiNet	0.8757	0.7952	0.2144	0.3875	0.3648
	DeepLabV3+	0.8673	0.7791	0.1306	0.5912	0.4424
Sichuan	BDGI-Net	0.7854	0.6307	0.1510	0.3614	0.3074
	HBGNet	0.7710	0.6301	0.1595	0.3730	0.3189
	TFNet	0.7607	0.5285	0.1684	0.4512	0.3885
	REAUNet	0.7295	0.5870	0.0706	0.6828	0.4985
	SEANet	0.7608	0.6242	0.1421	0.4027	0.3369
	BsiNet	0.7421	0.5915	0.1677	0.4127	0.3522
	DeepLabV3+	0.7239	0.6068	0.1345	0.5446	0.4253

boundary-aware, and multi-task decoding strategies.

Table 3 compares all methods across the Shandong plains, Netherlands polders, and Sichuan hilly region. BDGI-Net achieves the highest Precision, IoU, and lowest GTC in Shandong, with values of 0.9274, 0.8306, and 0.2445, respectively. This result indicates that the method preserves both region accuracy and long, regular boundaries. In the Netherlands, it reaches a Precision of 0.9049 and a GTC of 0.3284. HBGNet yields a slightly higher IoU (0.8241 versus 0.8225), whereas BDGI-Net provides higher Precision and lower GTC, giving a more favorable balance between semantic accuracy and parcel separation. In Sichuan, BDGI-Net obtains the highest IoU (0.6307) and lowest GTC (0.3074) under irregular parcel structures and complex terrain. Relative to the strongest competing result for each metric in each scene, BDGI-Net improves Precision by 0.42–1.44 percentage points and reduces GTC by 1.15–2.37 percentage points. Its IoU is best in Shandong and Sichuan and is 0.16 percentage points below HBGNet in the Netherlands. The improvement is therefore expressed more consistently in Precision and parcel-level geometry than in IoU alone. This distinction is relevant because high regional overlap does not necessarily

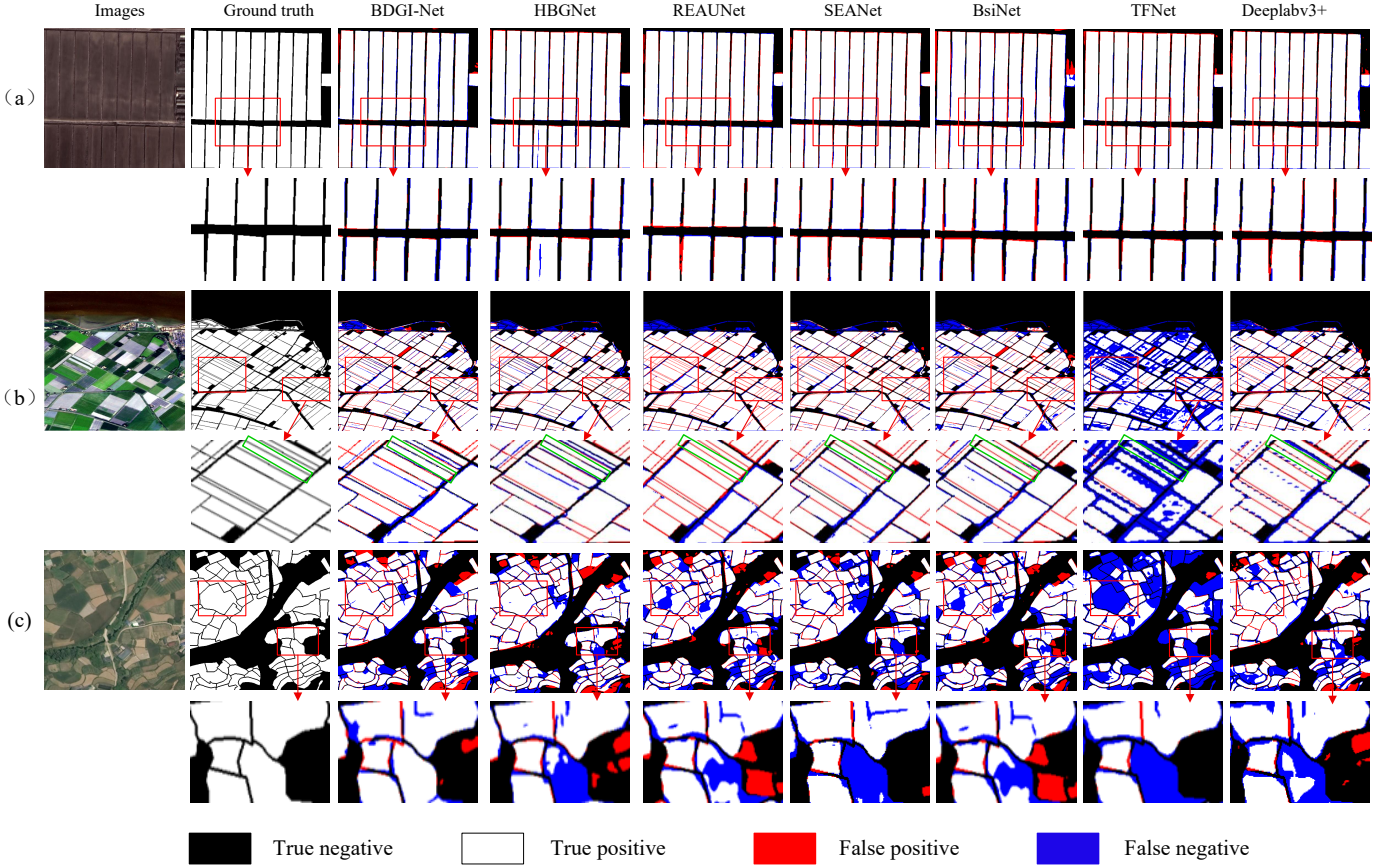


Figure 7: Error maps of cropland field parcel extraction results by the different models. White represents the correctly extracted range of cropland field parcels, black indicates the correctly extracted range of non-cropland field parcels, red signifies missed extraction areas, and blue denotes incorrect extraction areas. (a) Shandong Plain, (b) Netherlands Polder, and (c) Sichuan Hilly Region.

prevent parcel adhesion or omission.

Fig. 7 provides a visual comparison of different methods across the three study areas. In the Shandong plains, parcels are generally large and regular, but their long boundaries still pose challenges for maintaining continuous delineation. Compared with the other methods, BDGI-Net produces more complete parcel regions and preserves relatively clear and continuous boundaries. In the densely distributed Netherlands polders, adjacent parcels often exhibit similar spectral and textural characteristics, increasing the likelihood of boundary ambiguity and parcel adhesion. BDGI-Net alleviates these errors by incorporating explicit boundary guidance into semantic refinement, resulting in clearer separation between neighboring parcels. In the Sichuan hilly region, where parcels exhibit more irregular shapes, complex textures, and relatively weak contrast, BDGI-Net also preserves more complete parcel structures and reduces visually apparent omissions and boundary discontinuities.

Table 4 compares computational complexity. BDGI-Net uses 20.09M parameters and 321.98G FLOPs. TFNet and BsiNet are lighter, but their performance is less competitive in several challenging scenes. REAUNet and HBGNet achieve strong results at greater computational cost. For example, REAUNet requires 142.40M parameters and 2035.05G FLOPs, whereas HBGNet uses 30.45M parameters and 463.81G FLOPs. BDGI-

Net therefore occupies an intermediate complexity range while retaining favorable pixel-level and geometric performance. Its gains are not explained by model size alone, although BDGI-Net remains costlier than the lightest baselines.

Table 4: Comparison of computational complexity among different networks.

Network	Params (M)	FLOPs (G)
BDGI-Net (Ours)	20.09	321.98
HBGNet	30.45	463.81
TFNet	1.58	135.77
REAUNet	142.40	2035.05
SEANet	28.75	1666.94
BsiNet	7.85	198.67
DeepLabV3+	5.81	52.86

5.3. Descriptive Accuracy–Complexity Analysis

To connect scene-level accuracy with deployment cost, we derive an unweighted macro-average across the three test scenes in Table 3 and relate it to the parameter and FLOP counts in Table 4. This aggregation is descriptive: it summarizes the reported point estimates but does not pool pixels across datasets or replace repeated-run uncertainty. BDGI-Net attains the highest macro IoU and lowest macro GTC. Relative to HBGNet, which has nearly identical macro IoU, BDGI-Net reduces macro GTC

Table 5: Descriptive accuracy–complexity aggregation derived from Tables 3 and 4. Macro IoU and macro GTC are unweighted means across the Shandong, Netherlands, and Sichuan test scenes.

Network	Macro IoU↑	Macro GTC↓	Params (M)↓	FLOPs (G)↓
BDGI-Net (Ours)	0.7613	0.2934	20.09	321.98
HBGNet	0.7610	0.3152	30.45	463.81
TFNet	0.7102	0.4223	1.58	135.77
REANet	0.7370	0.3911	142.40	2035.05
SEANet	0.7525	0.3168	28.75	1666.94
BsiNet	0.7330	0.3361	7.85	198.67
DeepLabV3+	0.7291	0.3905	5.81	52.86

by 2.18 percentage points while using 34.0% fewer parameters and 30.6% fewer FLOPs. BDGI-Net requires 6.09× the FLOPs of the lightest baseline, DeepLabV3+, but improves macro IoU by 3.22 percentage points and reduces macro GTC by 9.71 percentage points. The proposed model therefore occupies the high-accuracy end of the observed trade-off rather than the lowest-compute regime.

5.4. Ablation Study

To determine which design choice produces each observed benefit, we conduct controlled removals on the same Netherlands setting used for task-weight selection. Single removals test the direct necessity of task-specific decoding, directional guidance, cross-dimensional reinforcement, and local–global context; paired removals test whether the corresponding effects are complementary rather than redundant. We retain the remaining training and evaluation conditions so that the metric and error-map changes can be linked to the removed mechanism.

Table 6: BDGI-Net variants. A check mark (✓) denotes a present module; an empty cell denotes removal.

Variant	GMRM	Bbranch	TAB	ETA
BDGI-Net	✓	✓	✓	✓
No_GMRM		✓	✓	✓
No_Bbranch	✓		✓	✓
No_TAB	✓	✓		✓
No_ETA	✓	✓	✓	
No_TAB_ETA	✓	✓		
No_Bbranch_ETA	✓		✓	
No_Bbranch_TAB	✓			✓
No_GMRM_ETA		✓	✓	
No_GMRM_TAB		✓		✓
No_GMRM_Bbranch			✓	✓

We construct ten variants to isolate the contributions of GMRM, the boundary decoder (Bbranch), ETA, and TAB, as summarized in Table 6. The four single-removal variants test each component directly, while six paired removals examine whether their effects remain complementary. Removing Bbranch replaces task-specific boundary decoding with a shared structure, whereas removing GMRM eliminates boundary-guided semantic refinement. The remaining variants remove ETA and/or TAB to assess hierarchical reinforcement and contextual modeling.

The Netherlands dataset offers a demanding test because its parcels are small, dense, and morphologically heterogeneous.

Table 7 shows that the complete model has the highest Precision (0.9049) and IoU (0.8225), along with the lowest GUC

Table 7: BDGI-Net ablation performance on the Netherlands dataset.

Variant	Precision↑	IoU↑	GOC↓	GUC↓	GTC↓
BDGI-Net	0.9049	0.8225	0.2555	0.2859	0.3284
No_GMRM	0.8845	0.8077	0.2408	0.3187	0.3429
No_Bbranch	0.8789	0.8086	0.2695	0.3043	0.3516
No_ETA	0.8899	0.8133	0.2538	0.3162	0.3454
No_TAB	0.8989	0.8094	0.2851	0.2904	0.3444
No_TAB_ETA	0.8830	0.8066	0.2226	0.3449	0.3605
No_Bbranch_ETA	0.8754	0.8036	0.2297	0.3384	0.3527
No_Bbranch_TAB	0.8789	0.8060	0.2058	0.3670	0.3592
No_GMRM_ETA	0.8794	0.8050	0.2129	0.3401	0.3529
No_GMRM_TAB	0.8817	0.8073	0.2088	0.3497	0.3555
No_GMRM_Bbranch	0.8740	0.8016	0.1282	0.4678	0.3789

(0.2859) and GTC (0.3284). Some variants reduce GOC, but none matches this balance. Each single removal decreases Precision and IoU and increases GTC, supporting component complementarity. Paired removals cause larger geometric errors. No_TAB_ETA increases GTC to 0.3605, indicating that removing both reinforcement mechanisms is more damaging than removing either one alone. Variants that remove Bbranch together with ETA or TAB also show lower Precision and higher GTC than the complete model. In particular, No_GMRM_Bbranch reaches the lowest GOC (0.1282) but the highest GUC (0.4678) and GTC (0.3789). This imbalance shows that one favorable error component does not imply reliable parcel separation and motivates joint evaluation with GTC.

To determine whether each paired removal merely adds the two corresponding single-removal effects or exposes a stronger dependency, we conduct a descriptive interaction audit on the supplied ablation point estimates. We orient both metrics so that positive values denote degradation: $D_{\text{IoU}}(v) = \text{IoU}_{\text{full}} - \text{IoU}_v$ and $D_{\text{GTC}}(v) = \text{GTC}_v - \text{GTC}_{\text{full}}$. For components i and j , we then compute $R_m(i, j) = D_m(-i, -j) - D_m(-i) - D_m(-j)$. A positive residual indicates supra-additive degradation in the reported setting, whereas a nonpositive residual indicates overlapping or sub-additive effects. Because the workbook contains aggregate point estimates rather than repeated runs, this audit diagnoses effect structure but is not a statistical interaction test.

Only the GMRM–Bbranch pair produces a positive GTC residual: removing both increases GTC by 5.05 percentage points, compared with the 3.77-point sum of their individual effects, leaving a +1.28-point residual. This pattern is consistent with the intended sequence in which Bbranch first reconstructs boundary-sensitive features and GMRM then converts that representation into semantic guidance. The remaining GTC residuals and all IoU residuals are nonpositive, so the data do not justify a claim of universal synergy among all modules. Instead,

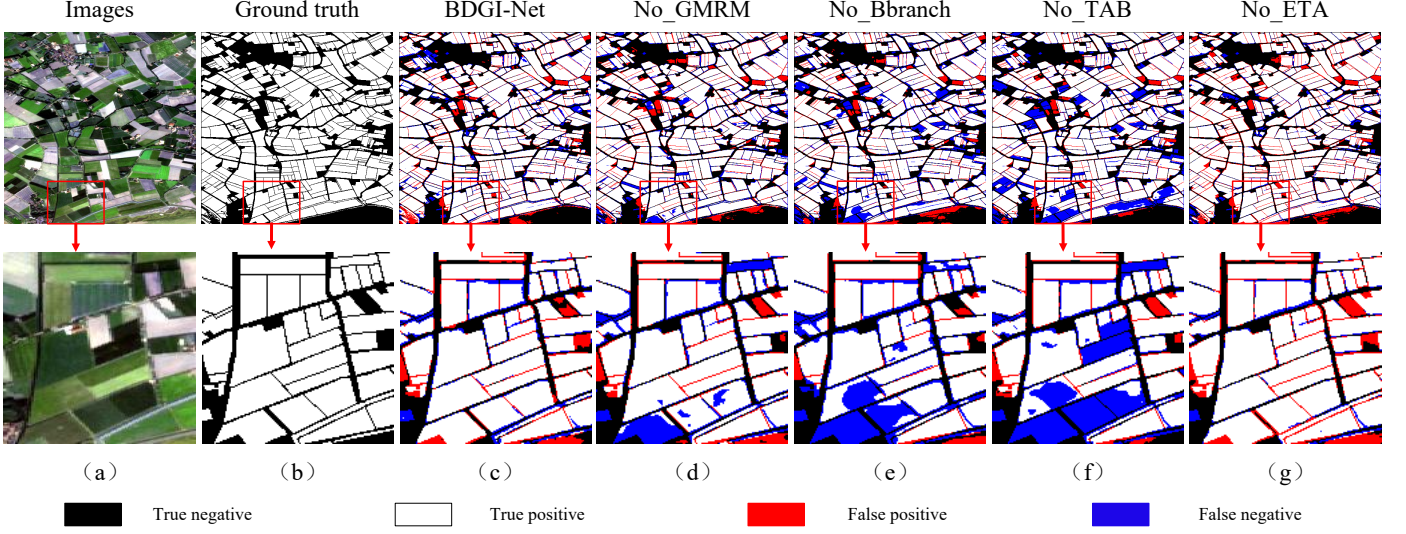


Figure 8: Cropland parcel error maps for BDGI-Net and its ablations. (a) Input image, (b) ground truth, (c) full-model segmentation, and (d)–(g) results after removing individual modules.

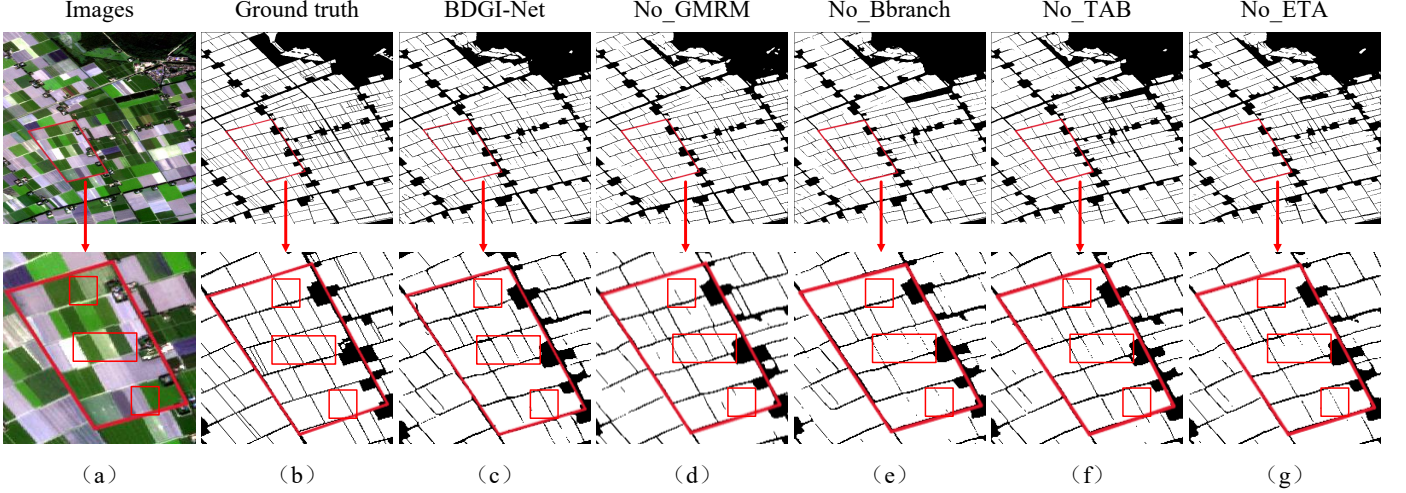


Figure 9: Cropland field parcel extraction results of BDGI-Net and its variants on the Netherlands dataset. (a) Original input image, (b) Ground truth mask with distinct boundaries, (c) Segmentation result of the full BDGI-Net, showing sharp separation between adjacent parcels, (d)–(g) Results of ablation variants.

Table 8: Descriptive pair-removal interaction audit on the Netherlands ablation results. All entries are percentage points. Δ denotes the observed degradation relative to the complete model, and R is the observed pair-removal degradation minus the sum of the two single-removal degradations.

Removed pair	$\Delta\text{IoU}\downarrow$	R_{IoU}	$\Delta\text{GTC}\uparrow$	R_{GTC}
GMRM + ETA	1.75	-0.65	2.45	-0.70
GMRM + TAB	1.52	-1.27	2.71	-0.34
GMRM + Bbranch	2.09	-0.78	5.05	+1.28
Bbranch + ETA	1.89	-0.42	2.43	-1.59
Bbranch + TAB	1.65	-1.05	3.08	-0.84
TAB + ETA	1.59	-0.64	3.21	-0.09

they identify the boundary-decoding–guided–interaction link as the clearest coupled relation in the reported ablation setting.

Figs. 9 and 8 further show how individual removals affect parcel regions and boundaries.

Effectiveness of ETA. We design ETA to stabilize parcel

representation through cross-dimensional interaction and global channel context. Removing it decreases Precision from 0.9049 to 0.8899 and IoU from 0.8225 to 0.8133, while increasing GTC from 0.3284 to 0.3454. Figs. 9(g) and 8(g) show greater intra-parcel fragmentation, more false negatives, and scattered background errors. Because the degradation occurs when both rotated interactions and their global complement are removed, the result supports ETA in suppressing locally unstable responses across heterogeneous parcel appearances.

Effectiveness of TAB.

TAB addresses the mismatch between local boundary detail and the long-range context required for complete parcel regions. Removing it decreases Precision from 0.9049 to 0.8989 and IoU from 0.8225 to 0.8094, while increasing GTC from 0.3284 to 0.3444. Figs. 9(f) and 8(f) show fragmented regions and more false negatives within large parcels. These changes support the

parallel Window MSA and Global MSA design, which preserves local structure while maintaining region-level continuity.

Effectiveness of Bbranch. Bbranch tests whether task-specific boundary reconstruction is necessary before guided interaction. Removing it decreases Precision from 0.9049 to 0.8789 and IoU from 0.8225 to 0.8086, while increasing GTC from 0.3284 to 0.3516. Figs. 9(e) and 8(e) show more boundary omissions and erroneous parcel connections. The simultaneous deterioration in region and geometric metrics supports separate decoding as a means of retaining boundary-sensitive features without forcing them into semantic reconstruction.

Effectiveness of GMRM.

GMRM tests whether the separated streams still require controlled information exchange. Removing it decreases Precision from 0.9049 to 0.8845 and IoU from 0.8225 to 0.8077, while increasing GTC from 0.3284 to 0.3429. Figs. 9(d) and 8(d) show more omissions, discontinuities, and erroneous connections near parcel boundaries. Because both decoders remain present in this variant, the comparison isolates the value of converting boundary features into spatial guidance for semantic refinement. This supports directional interaction as the complement to task-specific decoding, not unrestricted fusion.

5.5. Stability under Simulated Degradations

Image acquisition, transmission, and atmospheric interference can blur boundaries, reduce contrast, and corrupt parcel textures. To test whether the model preserves both regional overlap and parcel geometry when these observations deteriorate, we construct five degraded versions of the Netherlands test set by simulating blur, cloud occlusion, fog, Gaussian noise, and stripe noise [37, 38]. These tests evaluate stability under the specified synthetic corruptions rather than geographical or sensor-domain generalization.

Table 9 shows that BDGI-Net remains competitive across the five degradation types. Under blur, its IoU decreases by only 1.0 percentage point to 0.813, the highest value among the compared methods. It also achieves the highest IoU under fog (0.791), Gaussian noise (0.567), and stripe noise (0.767). Gaussian noise causes the largest decline, reducing IoU by 25.6 percentage points, but several competing methods deteriorate more severely. The GTC results are more mixed: BDGI-Net is best under blur, fog, and stripe noise, whereas HBGNet is slightly better under cloud occlusion and BsiNet under Gaussian noise. BDGI-Net is therefore not best on every metric or degradation type, but it generally maintains a favorable balance between segmentation accuracy and structural consistency.

A single-metric ranking cannot determine whether a model preserves overlap by sacrificing parcel separation, or vice versa. To evaluate this trade-off without imposing an arbitrary weighted score, we perform a joint dominance analysis over the supplied point estimates. For each original or degraded condition, model *A* dominates model *B* when *A* has no lower IoU and no higher GTC, with at least one strict inequality. We also report the mean and worst values across the five degraded conditions so that strong average behavior cannot conceal a brittle failure case.

Table 10: Descriptive multi-objective robustness summary derived from Table 9. Mean and worst values use the five degraded conditions; ND reports the number of original or degraded conditions in which a method belongs to the IoU–GTC Pareto-nondominated set. These aggregate point estimates do not provide repeated-run uncertainty.

Method	Mean IoU↑	Worst IoU↑	Mean GTC↓	Worst GTC↓	ND/6↑
BDGI-Net	0.7442	0.567	0.4166	0.514	5
HBGNet	0.7202	0.485	0.4348	0.532	2
REAU-Net	0.5642	0.078	0.5084	0.651	0
SEANet	0.6228	0.034	0.4970	0.597	1
BsiNet	0.6506	0.230	0.4458	0.496	1
DeepLabV3+	0.6548	0.269	0.5098	0.552	1
TFNet	0.6626	0.322	0.5352	0.573	1

BDGI-Net belongs to the nondominated set in five of the six conditions and dominates a competing method in 30 of the 36 condition–baseline pairs. Across the degraded conditions, it also achieves the highest mean IoU (0.7442), highest worst-case IoU (0.567), and lowest mean GTC (0.4166). BsiNet obtains a lower worst-case GTC (0.496 versus 0.514), so the analysis does not imply that BDGI-Net is uniformly best. Instead, it shows that BDGI-Net most consistently occupies the favorable region of the joint accuracy–geometry trade-off represented by the available point estimates.

Fig. 10 provides representative segmentation results for the five degradation types. Under fog and cloud occlusion, reduced contrast obscures portions of parcel boundaries, yet BDGI-Net generally preserves coherent regions. Blur weakens fine edges but causes relatively limited structural change. Gaussian and stripe noise introduce stronger local fragmentation and boundary errors because they directly corrupt spatial patterns. Even under these conditions, many parcels retain their overall organization, which is consistent with the relative trends in IoU and GTC.

Figs. 11 and 12 summarize compound-degradation performance. BDGI-Net retains competitive IoU as severity increases and generally produces lower parcel-level geometric errors than the baselines. IoU measures regional overlap, whereas GTC reveals whether degraded predictions preserve parcel geometry. Although all methods deteriorate with image quality, BDGI-Net retains coherent regions and separable structures over the tested range. These results support relative robustness under the simulated conditions, not general robustness across all naturally occurring image degradations.

Table 9: IoU $\uparrow$ (top) and GTC $\downarrow$ (bottom) of seven networks on the Netherlands dataset under five degradation conditions. Parenthetical values are absolute changes in percentage points relative to the original images, calculated as (current – original) $\times$ 100. The best metric value in each row is highlighted in bold; for each degradation condition, the smallest absolute change relative to the original setting is also highlighted in bold.

IoU $\uparrow$	BDGI-Net(ours)	HBGNet'25	REANet'24	SEANet'23	BsiNet'22	DeepLabV3+'18	TFNet'24
Original	0.823	0.824	0.801	0.813	0.795	0.779	0.776
Blur	0.813 ($\downarrow 1.0$ pp)	0.801 ($\downarrow 2.3$ pp)	0.723 ($\downarrow 7.8$ pp)	0.788 ($\downarrow 2.5$ pp)	0.780 ($\downarrow 1.5$ pp)	0.765 ($\downarrow 1.4$ pp)	0.740 ($\downarrow 3.6$ pp)
Cloud Occlusion	0.783 ($\downarrow 4.0$ pp)	0.785 ($\downarrow 3.9$ pp)	0.699 ($\downarrow 10.2$ pp)	0.790 ($\downarrow 2.3$ pp)	0.767 ($\downarrow 2.8$ pp)	0.758 ($\downarrow 2.1$ pp)	0.766 ($\downarrow 1.0$ pp)
Fog	0.791 ($\downarrow 3.2$ pp)	0.772 ($\downarrow 5.2$ pp)	0.646 ($\downarrow 15.5$ pp)	0.750 ($\downarrow 6.3$ pp)	0.741 ($\downarrow 5.4$ pp)	0.757 ($\downarrow 2.2$ pp)	0.747 ($\downarrow 2.9$ pp)
Gaussian Noise	0.567 ($\downarrow 25.6$ pp)	0.485 ($\downarrow 33.9$ pp)	0.078 ($\downarrow 72.3$ pp)	0.034 ($\downarrow 77.9$ pp)	0.230 ($\downarrow 56.5$ pp)	0.269 ($\downarrow 51.0$ pp)	0.322 ($\downarrow 45.4$ pp)
Stripe Noise	0.767 ($\downarrow 5.6$ pp)	0.758 ($\downarrow 6.6$ pp)	0.675 ($\downarrow 12.6$ pp)	0.752 ($\downarrow 6.1$ pp)	0.735 ($\downarrow 6.0$ pp)	0.725 ($\downarrow 5.4$ pp)	0.738 ($\downarrow 3.8$ pp)
GTC $\downarrow$	BDGI-Net(ours)	HBGNet'25	REANet'24	SEANet'23	BsiNet'22	DeepLabV3+'18	TFNet'24
Original	0.328	0.354	0.353	0.352	0.365	0.442	0.475
Blur	0.365 ($\uparrow 3.7$ pp)	0.395 ($\uparrow 4.1$ pp)	0.429 ($\uparrow 7.6$ pp)	0.455 ($\uparrow 10.3$ pp)	0.398 ($\uparrow 3.3$ pp)	0.525 ($\uparrow 8.3$ pp)	0.546 ($\uparrow 7.1$ pp)
Cloud Occlusion	0.396 ($\uparrow 6.8$ pp)	0.391 ($\uparrow 3.7$ pp)	0.462 ($\uparrow 10.9$ pp)	0.445 ($\uparrow 9.3$ pp)	0.417 ($\uparrow 5.2$ pp)	0.516 ($\uparrow 7.4$ pp)	0.547 ($\uparrow 7.2$ pp)
Fog	0.388 ($\uparrow 6.0$ pp)	0.421 ($\uparrow 6.7$ pp)	0.496 ($\uparrow 14.3$ pp)	0.566 ($\uparrow 21.4$ pp)	0.484 ($\uparrow 11.9$ pp)	0.552 ($\uparrow 11.0$ pp)	0.573 ($\uparrow 9.8$ pp)
Gaussian Noise	0.514 ($\uparrow 18.6$ pp)	0.532 ($\uparrow 17.8$ pp)	0.651 ($\uparrow 29.8$ pp)	0.597 ($\uparrow 24.5$ pp)	0.496 ($\uparrow 13.1$ pp)	0.504 ($\uparrow 6.2$ pp)	0.513 ($\uparrow 3.8$ pp)
Stripe Noise	0.420 ($\uparrow 9.2$ pp)	0.435 ($\uparrow 8.1$ pp)	0.504 ($\uparrow 15.1$ pp)	0.422 ($\uparrow 7.0$ pp)	0.434 ($\uparrow 6.9$ pp)	0.452 ($\uparrow 1.0$ pp)	0.497 ($\uparrow 2.2$ pp)

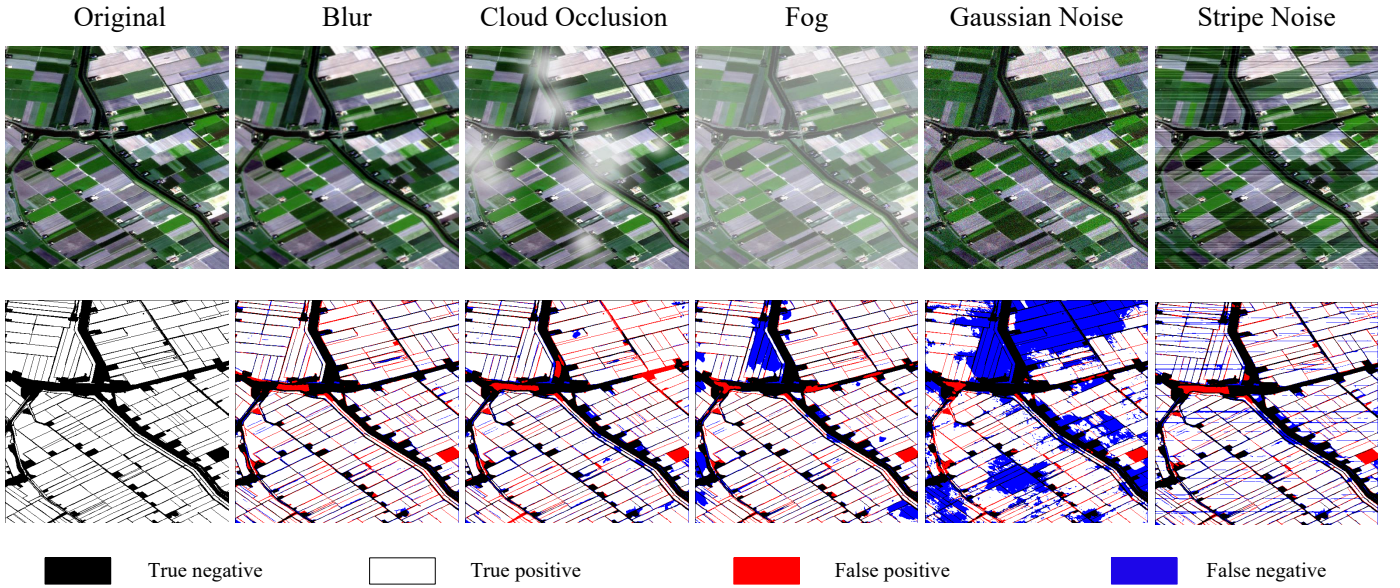


Figure 10: Visual examples of the five simulated degradation conditions on the Netherlands dataset. The top row shows the degraded input images, while the bottom row presents the corresponding segmentation results produced by BDGI-Net.

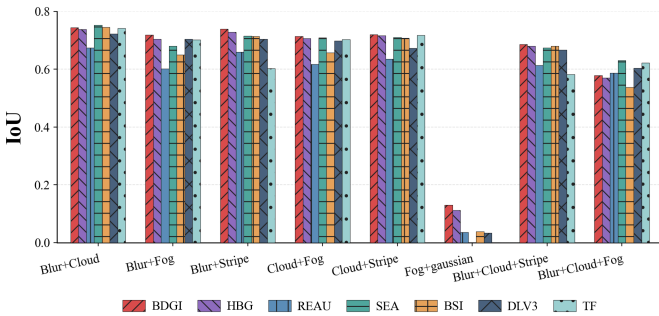


Figure 11: Comparison of IoU metrics on the Netherlands dataset.

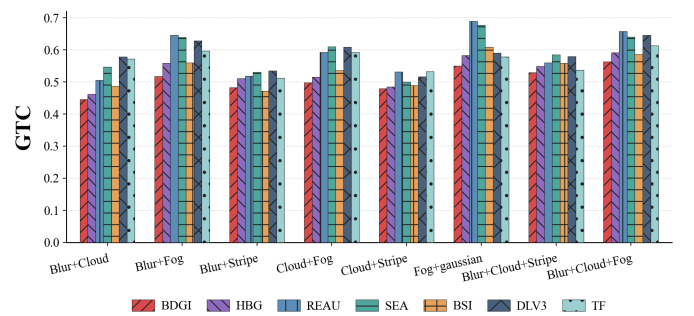


Figure 12: Comparison of GTC metrics on the Netherlands dataset.

6. Discussion

The results clarify why parcel boundaries should be treated as controlled auxiliary information rather than as either an implicit cue or a fully shared prediction objective. The separate decoding streams preserve the different reconstruction requirements of

parcel interiors and boundary transitions, while GMRM restores only the directional interaction needed to refine semantic masks. The ablation results support this ordered relation: removing either task-specific boundary reconstruction or guided interaction weakens the balance between regional overlap and parcel-level geometry, and their joint removal produces a +1.28-percentage-

point supra-additive GTC degradation in the supplied point estimates. Because the other pairwise residuals are nonpositive, this evidence supports the specific Bbranch–GMRM dependency rather than a generic synergy claim for every module. The comparison across the three scenes further shows that the principal benefit is not an unconditional increase in every pixel metric. Instead, BDGI-Net more consistently reduces parcel-level total classification error while remaining competitive in IoU. This distinction is important for applications in which a small overlap gain cannot compensate for merging adjacent management units.

Broader operational use changes the next technical question from delineating individual scenes to maintaining reliable interaction across sensors, seasons, and acquisition conditions. The present evaluation covers GF-2 and Sentinel-2 observations from three regions and specified synthetic degradations; it does not quantify leave-one-region-out transfer, temporal drift, or naturally paired corruption. As mapping expands, boundary confidence may vary more strongly than semantic confidence, so a fixed boundary-to-semantic interaction may no longer be optimal for every acquisition. To preserve semantic ownership under this shift, we will extend GMRM with a confidence-bounded reliability gate: semantic uncertainty will define where correction is permissible, calibrated boundary confidence will quantify whether the auxiliary evidence is trustworthy, and local structural compatibility will determine whether the boundary cue matches the image. The gate will remain zero when any condition fails. We will evaluate this extension by holding out sensors, regions, and acquisition periods and jointly reporting calibration, IoU, and parcel-level geometric error. Lightweight sensor adapters and cross-sensor self-supervised alignment can then address residual domain shift without replacing the task-specific decoders.

Scaling the delineation workflow also increases the demand for boundary annotation. Dense parcel and boundary labels are practical for controlled model development but become costly when new regions and seasons are added. The dual-stream structure provides a direct route to label-efficient expansion: a teacher model can generate separate region and boundary predictions, topology-aware filtering can retain pseudo-labels only when the two outputs form a consistent parcel, and active sampling can prioritize boundary bands with high disagreement. This design preserves the distinction between the two objectives rather than collapsing them into one pseudo-label. We will evaluate its value through performance-versus-label curves, annotation time, pseudo-boundary precision, and the stability of GTC under decreasing label fractions [39].

More accurate parcel separation also makes downstream spatial products feasible at a larger scale, which shifts attention from raster masks to persistent parcel objects. A practical next step is to couple BDGI-Net with graph-constrained polygonization and temporal parcel matching. The segmentation output would provide candidate regions and boundaries; a topology layer would enforce non-overlap and closed polygons; and temporal matching would maintain parcel identities across acquisitions while recording split and merge events. Uncertainty from the semantic and boundary streams should be propagated to the resulting

polygons rather than discarded during vectorization. Evaluation should therefore include topology validity, polygon displacement, temporal identity consistency, and end-to-end effects on parcel-level area or change estimates. This would test whether the information interaction established in the present study can support a complete mapping workflow without overstating what the current raster experiments demonstrate.

7. Conclusion

This study addresses the tension between regional coherence and boundary discrimination in heterogeneous parcel segmentation. We develop BDGI-Net with task-specific semantic and boundary decoders and directional GMRM boundary guidance; ETA and TAB further reinforce cross-dimensional responses and local–global context. Across GF-2 and Sentinel-2 data from the Shandong plains, Netherlands polders, and Sichuan hilly region, BDGI-Net improves Precision by 0.42–1.44 percentage points and reduces GTC by 1.15–2.37 percentage points relative to the strongest result for each scene and metric. It achieves the best IoU in two scenes and remains within 0.16 percentage points of the best result in the Netherlands. Ablation and synthetic-degradation tests support the components’ complementary roles and stability within the tested conditions, while a joint robustness analysis places BDGI-Net in the Pareto-nondominated set in five of six original or degraded conditions. The results support task-specific decoding followed by controlled interaction as a practical way to preserve parcel regions and geometric separation. Future work will pursue reliability-calibrated cross-sensor interaction, label-efficient dual-stream learning, and topology-aware temporal parcel mapping.

CRedit authorship contribution statement

Mei Yang: Conceptualization, Supervision, Funding acquisition, Writing – review & editing. Bo Yu: Methodology, Software, Experiments, Data curation, Writing – original draft. Chong Zhang: Supervision, Methodology, Funding acquisition, Experiments, Writing – review & editing. Zhen Pan: Validation, Data curation. Lijing Gao: Validation, Funding acquisition. Fan Min: Supervision, Funding acquisition.

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The datasets used in this study are available from the corresponding author upon reasonable request.

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